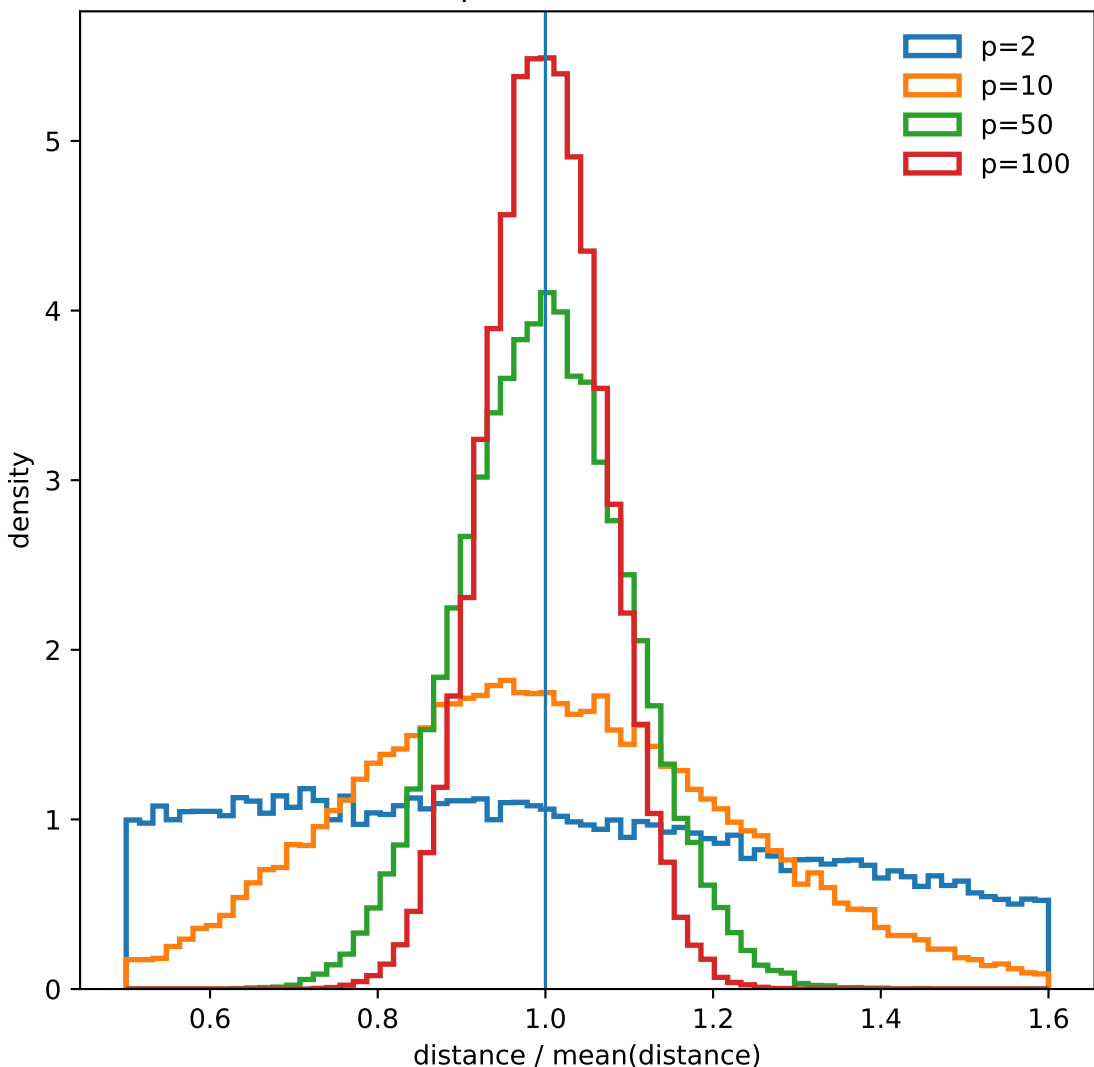


A. Normalized pairwise distance distributions



B. Distance concentration with dimension

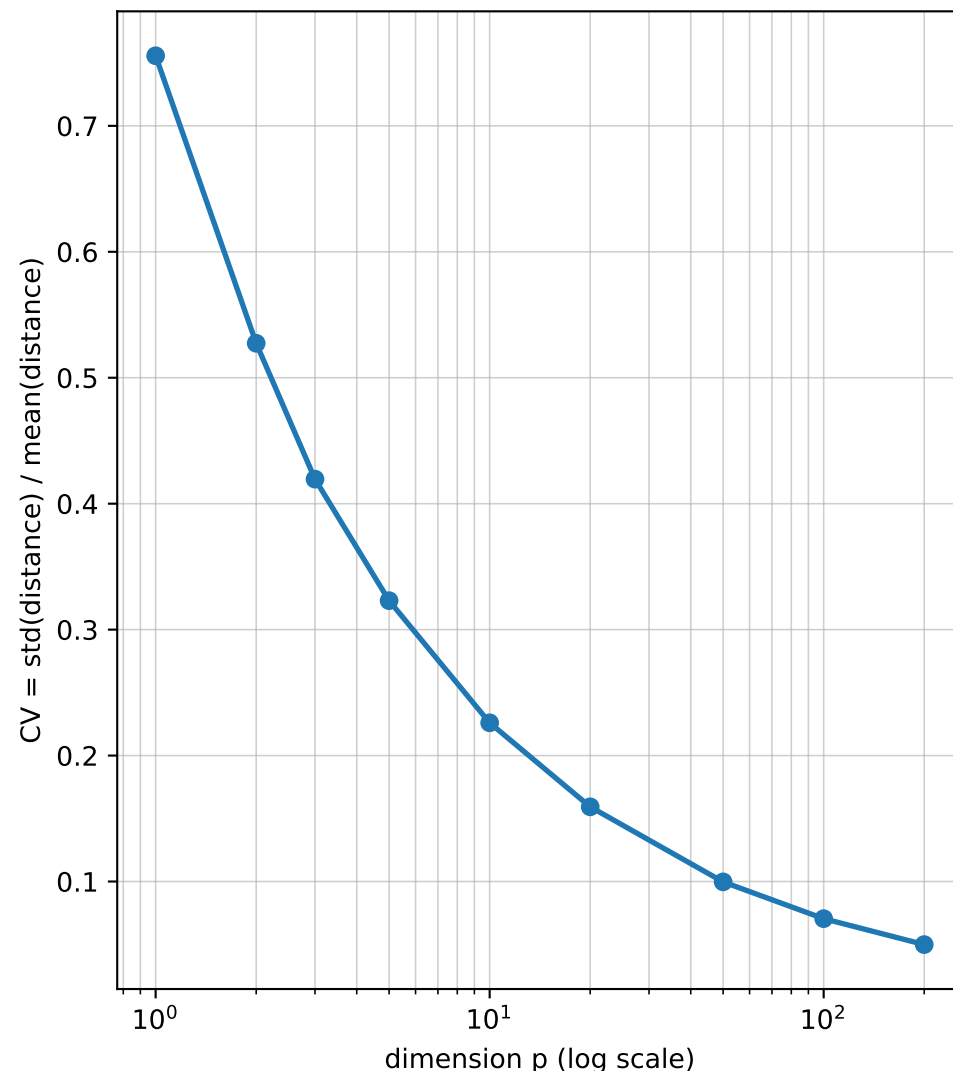


Figure: Demonstration of the curse of dimensionality via distance concentration. For each dimension p , random points are sampled from a p -dimensional standard normal distribution and Euclidean distances are computed between many random pairs. (A) After normalizing by the mean distance, the distance distributions cluster more tightly around 1 as p increases, showing that pairwise distances become more similar. (B) The coefficient of variation ($CV = \text{std}(\text{distance}) / \text{mean}(\text{distance})$) decreases with p , summarizing this convergence. Takeaway: in high dimensions, distances between random points become less variable, so distance is less useful for distinguishing which points are truly 'close' versus 'far'.